ThaliSense

Eat the food you love.
Know what it does to you.

An on-device AI nutrition and metabolic-health coach built for Indian meals.

NEXUS InnovateX 2026 · IdeaPitch

Live: multihacks-unstop26.vercel.app

github.com/andringodson/Hackathon-MultiHacksUnstop26

PROBLEM

India’s diabetes and obesity crisis is cooked in everyday kitchens, and our tools don’t speak its language.

101 M
Indians living with diabetes

136 M
more with prediabetes, mostly undiagnosed

BMI 23
is where overweight starts for Asian Indians, not the global 25

- Calorie apps are built for Western diets. They don't know pesarattu, bisi bele bath or kadhi, or what “do roti aur dahi” means.
- Risk is mis-measured. South Asians get diabetes at lower weight, so global BMI tells at-risk people they're fine.
- “Eat less rice” doesn't stick, dietitians are unaffordable, and people won't upload health data to a server.

Source: ICMR-INDIAB, Lancet Diabetes & Endocrinology 2023; Misra et al., JAPI 2009

TARGET USERS

Built for the Indian family kitchen, not the gym.

Adults at metabolic risk

25 to 55, desk jobs, family history of diabetes, a growing waistline. They want to act early without giving up home food.

People with prediabetes

Told by a doctor to “watch your diet”, with no idea what that means for idli, roti or biryani.

Home cooks and caregivers

The person planning meals for the whole family, on a budget, in a regional cuisine.

Later: health workers

ASHA and community health workers screening villages with the Indian Diabetes Risk Score, offline on one phone.

Later: employers and insurers

Corporate wellness and health-insurance programmes that need a private, low-cost prevention tool.

PROPOSED SOLUTION

A coach that understands Indian food, measures risk the Indian way and plans meals you'll actually eat.

1 · Log

Type it your way ("rendu dosai", "2 roti, dal") or snap a photo. On-device AI recognises 108 Indian dishes.

2 · Understand

Indian Diabetes Risk Score, Asian BMI cut-offs, waist-to-height ratio and personal calorie and protein targets.

3 · Coach

An agent finds what matters most and swaps foods you already eat, showing the weekly impact of each swap.

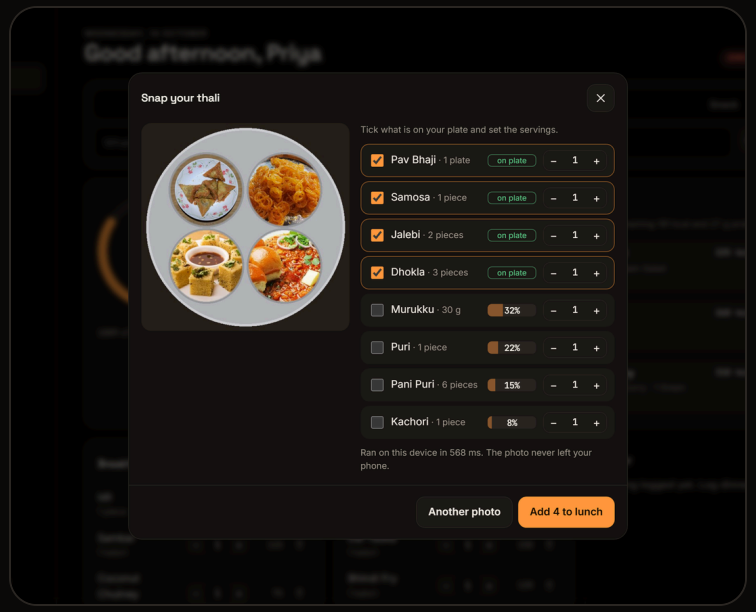
4 · Plan

A self-checked 7-day regional meal plan for your diet and ₹/day budget. Log any meal with one tap.

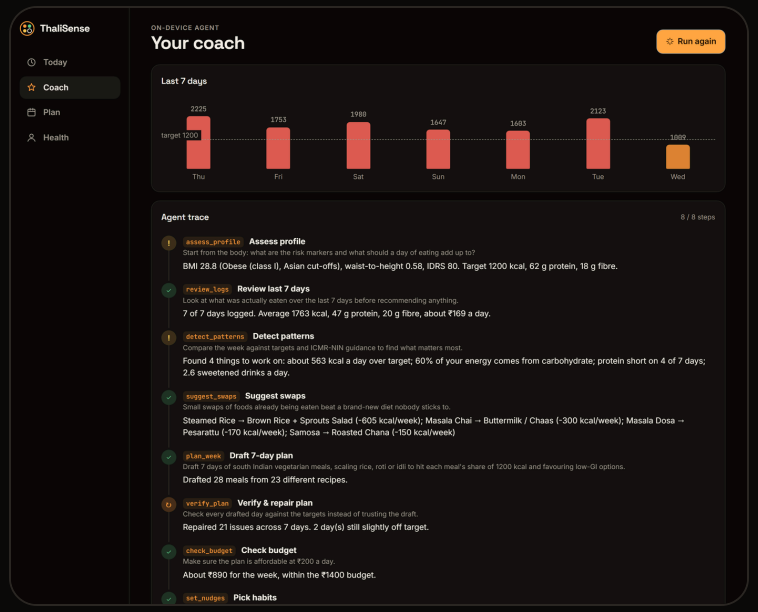
Free. No sign-up. Nothing leaves your phone: the AI runs inside the browser.

WORKING MVP · LIVE TODAY

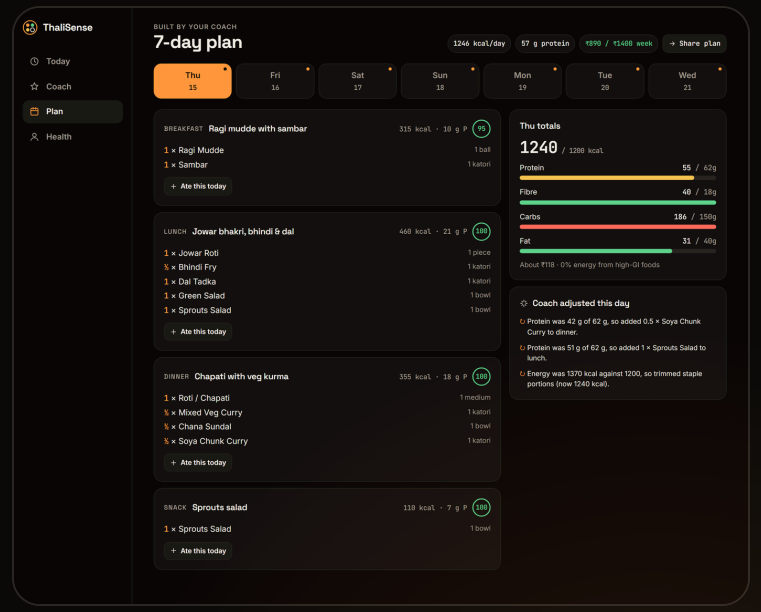
From a photo of your thali to a week of better meals.



Snap a whole thali: the AI scans the plate and ticks every dish it finds, on the device.



Coach: an 8-step agent trace you can follow, plus insights and swaps.



Plan: 7 regional days within budget, with notes on what the coach adjusted.

Try the demo week at multihacks-unstop26.vercel.app · Thali photo: composite of Wikimedia Commons images, CC BY-SA 4.0 (credits in the repo)

ThaliSense

KEY FEATURES

Everything a dietitian would do in the first week, for free.

Indian-first food data

108 dishes with IFCT-2017-based nutrition, GI band and rupee cost; Hindi, Tamil and regional names.

Natural-language logging

Quantities, “do / rendu / aadha”, plurals, typos and compound dishes like curd rice.

Photo recognition

Fine-tuned on-device vision model: 87.8% top-1, 95% top-3 on 1,694 photos, and it reads whole thalis.

Clinical risk scores

MDRF Indian Diabetes Risk Score, Asian BMI, waist-to-height ratio, with a prompt to get tested if at high risk.

Explainable agent

Shows each step it takes, swaps foods with measured weekly impact, and verifies and repairs its own plan.

Private, offline, installable

No account and no server; installs like an app, works offline, and costs nothing to run per user.

TECHNOLOGY · AI USED

Real AI, running entirely on the user's phone.

input
Typed meal / photo
→
parser · MobileCLIP-S2
Dish recognition
→
localStorage
Meal log
→
coach agent
8 tool steps
→
output
Insights · swaps · plan

Vision: fine-tuned MobileCLIP on the phone

Transformers.js on ONNX Runtime Web, WebGPU or 4-thread WASM, in a Web Worker. One vector per dish from descriptive prompts, few-shot adapted on 2,354 labelled photos. The 72 MB encoder downloads once; a second pass scans five plate regions to find every dish on a thali.

Agent: plan, verify, repair

A tool-based planner: `assess → review → detect → swap → plan → verify → budget → habits`. It scores 60 real meal templates on protein, fibre, glycaemic load, cost and region, scales the staples, then checks every day against the targets and fixes the misses.

React 19 · TypeScript · Vite · Transformers.js · ONNX Runtime Web · Vitest · Playwright + axe · GitHub Actions · Vercel · PWA · no backend, no paid APIs

MEASURED, NOT GUESSED

We benchmarked four vision models and fine-tuned the best.

MODEL, ON 1,694 TEST PHOTOS	SIZE	TOP-1	TOP-3
CLIP ViT-B/32 (our v1)	89 MB	71.5%	86.8%
CLIP ViT-B/16	87 MB	73.6%	87.9%
MobileCLIP-S2	72 MB	81.5%	92.9%
MobileCLIP-S2, few-shot adapted	**72 MB**	**87.8%**	**95.0%**

25% → 55%

of dishes found on 200 four-dish test thalis, by adding a scan of five plate regions to the whole-photo guess.

Test photos from three public Indian-food datasets, never used for training or tuning. Full model card: github.com/andringodson/Hackathon-MultiHacksUnstop26/blob/main/docs/vision.md

MARKET & USE CASES

Prevention is India’s biggest health opportunity, and food is where it starts.

237 M

Indians with diabetes or prediabetes who would benefit from diet guidance today (ICMR-INDIAB 2023).

₹0

marginal cost per user: compute runs on the user's own phone, so the app scales without server bills.

4

regional cuisines and 3 diet types from day one, so it fits most Indian households.

Individuals

Daily logging, weekly coaching and a meal plan for the family.

Clinics & labs

Patients who score high on IDRS are referred for HbA1c testing; dietitians get a coaching companion app.

Public health

ASHA workers run IDRS screening offline as part of national NCD prevention efforts.

BUSINESS MODEL

Free for people. Paid by those who benefit from healthier people.

Free core app

Logging, risk scores, the coach and plans stay free forever. Because there are no server costs, free is sustainable.

B2B wellness licences

Employers and health insurers pay per member per year for a white-label version with opt-in, anonymised progress reports.

Diagnostic referrals

Users who score high on IDRS can book a fasting-glucose or HbA1c test with partner labs; we earn a referral fee.

Pro for dietitians

A subscription dashboard so dietitians and clinics can give clients ThaliSense plans that they review and adjust.

COMPETITIVE ADVANTAGE

Why ThaliSense wins where general calorie apps don’t.

	TYPICAL CALORIE APP	PAID DIET COACHING	THALISENSE
Understands Indian dishes & Hinglish	Partly	Yes	**Yes, 108 dishes, regional names**
Indian risk science (IDRS, BMI 23)	No	Sometimes	**Built in**
Explains its reasoning	No	Human coach	**Visible agent trace**
Data stays on your device	No, cloud	No, cloud	**Yes, nothing uploaded**
Cost to the user	Freemium	₹ thousands a month	**Free**
Budget-aware regional plans	No	Manual	**Automatic, self-checked**

EXPECTED IMPACT

Small swaps, caught early, at national scale.

~1,200

kcal a week saved by the swaps the coach finds in the demo week, about 0.7 kg of fat a month, with more protein and fibre.

60+

IDRS score triggers a clear “get tested” prompt, catching undiagnosed diabetes earlier.

0

bytes of health data uploaded. Privacy builds trust, especially for women and older users.

- Fights obesity with the food people already eat, not a foreign diet they'll quit in a week.
- Inclusive by design: vegetarian-aware, budget-aware (₹200 a day by default), regional, and runs on low-end phones.
- Scales at near-zero cost: static hosting, on-device AI, works offline.

FUTURE SCOPE

Where ThaliSense goes next.

- Plate segmentation: outline each katori, then recognise it, for crowded thalis.
- Voice logging in Hindi, Tamil, Bengali and Marathi.
- More dishes adapted: labelled photos for all 108 dishes, then the next 200.
- ASHA-worker mode: offline community IDRS screening and referrals.
- Wearable and glucometer import for steps and glucose response.
- Optional LLM chat, grounded in the agent's trace, with the user's own key.

ThaliSense

Try it now.

Open the link on any phone and press **Try with a demo week**.

Live app

multihacks-unstop26.vercel.app

Source code

github.com/andringodson/Hackathon-MultiHacksUnstop26